MILESTONE 4: Deployment

Activity 4.1: Run Locally

Run locally with:

Navigate to: http://127.0.0.1:8000 /html>

(base) telladivya...@...MacBook-Air edugenie
% uvicorn main:app --reload

INFO: Will watch for changes in these directories: ['/Users/telladivya.../Desktop/edugenie']

INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)

INFO: Started reloader process [17754] using StatReload

/opt/anaconda3/lib/python3.11/site-packages/torch/utils/dl...py:185: FutureWarning: optree is installed but the version is too old to

support PyTorch Dynamo in C++ pytrees. C++ pytrees support is disabled. Please consider

upgrading optree using 'python3 -m pip install

--upgrade 'optree>=0.13.0".

warnings.warn(

INFO: Started server process [17751]

INFO: Waiting for application startup.

INFO: Application startup complete.

Fig. Showing the Application can be accessed

Activity 4.2: Functional Testing

Ask a question

Get an explanation

Generate a quiz

Summarize content

Get a personalized learning planWelcome to EduGenie

Your personal AI tutor for learning support!

Ask EduGenie a Question:

[Which is the largest ocean?] [Ask EduGenie]

Need an Explanation?

[Process of Photosynthesis] [Explain]

Summarize a Paragraph:

[Enter a paragraph to summarize...]
[Summarize]

Fig. EDUGENIESummarize a Paragraph:

Fig. EDUGENIE

Summarize a Paragraph:

Enter a paragraph to summarize...
[Summarize]

Generate a Quiz:

Enter Topic: [Generate Quiz]

Get Learning Recommendations:

E.g., Linear Algebra... [GET RECOMMENDATION]

Fig. EDUGENIE

Exploring EduGenie:

a. Asking questions:

Welcome to EduGenie

Your personal AI tutor for learning support!

Ask EduGenie a Question:

which is the largest ocean? [Ask EduGenie]

Answer:

The Pacific Ocean is the largest ocean on Earth.

(Kela add moolama maraikappatta vari)

b. Explanation on any topic:

a. Explanation of any topic:

Need an Explanation?

Quantum Computing

Explanation:

Quantum computing is a type of computing that uses the principles of quantum

mechanics to perform certain calculations much faster than traditional computers. It allows computers to perform certain calculations much faster than traditional computers because it uses tiny particles called qubits that can exist in multiple states at the same time. This allows for new kinds of calculations and solving problems that are not possible with traditional computers.

Need an Explanation?

Binary Search Algorithm

Explanation: The binary search algorithm is a way to find the index of a target value in a sorted list of numbers. It's like searching for a needle in a haystack. You need to find the number between 0 and 1, and then go back to the beginning of the list to find the target value.

a. Summarising long paragraphs: Summarize a Paragraph:

The Industrial Revolution, which began in the late 18th century...

Summary:

The Industrial Revolution changed the world by shifting it from farming to factories. New

machines, like James Watt's steam engine, made things faster and easier to produce. While this boosted economies, it also created problems like bad working conditions, child labor, pollution, and overcrowded cities. Even with these problems, the Industrial Revolution shaped the modern world, affecting how we travel, communicate, work, and trade.

a. Generating quizzes:

It generates three questions with 4 options each. It corrects the answer if chosen wrong.

Generate a Quiz:

Pythagoras theorem

Quiz: Q1: What does the Pythagorean theorem describe?

The relationship between the angles of a triangle.

The relationship between the sides of a right-angled triangle.

The relationship between the area and perimeter of a triangle.

The relationship between the sides of any triangle. Check Answer

Quiz:

Q1: What does the Pythagorean theorem describe?

The relationship between the angles of a triangle.

The relationship between the sides of a right-angled triangle.

The relationship between the area and perimeter of a triangle.

The relationship between the sides of any triangle.

Check Answer

Correct!

Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c' is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

$a^2 + b^2 = c^2$

Q2: If 'a' and 'b' are the lengths of the two shorter sides of a right-angled triangle, and 'c'

is the length of the longest side (hypotenuse), what equation represents the Pythagorean theorem?

$a^2 + b^2 = c^2$

Q3: Which type of triangle does the Pythagorean theorem apply to?

Equilateral triangles

Isosceles triangles

Right-angled triangles

All types of triangles

a. Guiding with learning recommendations from beginners level to advanced level

Get Learning Recommendations:

SQL

Get Recommendations

Learning Recommendations for "SQL":

SQL Learning Path: From Zero to Hero

This learning path is structured to progressively introduce SQL concepts, starting

from the basics and gradually advancing to more complex topics. It's designed to be adaptive: feel free to adjust the pace and delve deeper into areas that particularly interest you.

1. Beginner Level: Building a Foundation

Estimated Time: 1–2 weeks

Key Topics:

What is a Database and SQL? (Relational Model, DBMS)

Basic Syntax (SELECT, FROM, WHERE)

Data Types (INT, VARCHAR, DATE, etc.)

Filtering Data (WHERE clause with comparison operators, logical operators: AND, OR, NOT)

Ordering Results (ORDER BY)

Limiting Results (LIMIT/OFFSET or TOP)

Basic Aggregate Functions (COUNT, SUM, AVG, MIN, MAX)

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I. Beginner Level: Building a Foundation

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Introduction to Tables and Columns

Resources:

Interactive Tutorials:

SQLZoo: Excellent for hands-on practice with different SQL flavors.

Khan Academy's SQL course: Offers a good introductory overview.

Codecademy's Learn SQL: Interactive lessons with practical exercises.

Videos:

FreeCodeCamp's SQL Tutorial: Comprehensive and beginner-friendly.

II. Intermediate Level: Working with Multiple Tables

(Estimated Time: 2–3 weeks)

Key Topics:

Joining Tables (INNER JOIN, LEFT JOIN, RIGHT JOIN, FULL OUTER JOIN)

Subqueries (Nested queries)

Grouping Data (GROUP BY)

Filtering Groups (HAVING)

Set Operations (UNION, INTERSECT, EXCEPT)

Working with Strings (string concatenation, substrings, LIKE operator)

Working with Dates and Times (date formatting, date arithmetic)

Views (Creating and using views)

Resources:

Books:

"SQL Queries for Mere Mortals" by Michael J. Hernandez: A great resource for understanding practical SQL querying.

Online Documentation: Refer to the documentation for your specific SQL database (e.g., MySQL, PostgreSQL, SQL Server).

Practice Platforms: LeetCode, HackerRank, and similar platforms offer SQL challenges to reinforce your skills.

III. Advanced Level: Mastering Database Management

(Estimated Time: 3–4 weeks and beyond)

Key Topics:

Stored Procedures and Functions

Triggers

Indexes and Performance Tuning

Transactions and Concurrency Control

Database Design (Normalization)

Books:

"SQL Performance Explained" by Markus Winand: In-depth coverage of SQL performance optimization.

"Database Internals" by Alex Petrov: Explains the inner workings of database systems.

Advanced Online Courses: Look for specialized courses on database administration and performance tuning on platforms like Udemy and Coursera.

Community Forums and Blogs: Engage with the SQL community to learn from experienced professionals and stay up-to-date with the latest advancements.

Adaptive Learning Tips:

Start with the basics: Don't rush into advanced topics before mastering the fundamentals.

Practice regularly: The more you practice writing SQL queries, the better you'll become.

Use real-world datasets: Working with real data will help you understand how SQL is used in practical scenarios.

Focus on understanding the concepts: Don't just memorize syntax; understand the underlying logic behind SQL queries.

Break down complex problems: Divide large problems into smaller, manageable parts.

Don't be afraid to experiment: Try different approaches and see what works best.

Seek help when needed: Don't hesitate to ask for assistance from online communities, forums, or mentors.

This learning path provides a solid foundation for mastering SQL. Remember to adapt it to your own learning style and goals, and enjoy the journey of becoming proficient in this powerful language!

It provides along with details of resources from where to learn and stepwise guidance throughout.